

VIKRAM CHENNAMSETTY

+91 9347636338 | Email | LinkedIn | GitHub | Portfolio

Professional Summary

Final-year B.Tech ECE student specialising in Signal Processing and Communication, with practical experience in Python scripting, IoT automation, ESP32/ESP32-CAM development, and hardware-integrated applications. Familiar with LoRa communication, KiCad PCB design basics, and MATLAB/Simulink workflows. Completed an IoT internship at Emertxe Information Technologies.

Education

B.Tech, Electronics and Communication Engineering	2023 – 2027
Specialisation: Signal Processing	CGPA: 8.50 (up to 5th Semester)
Kalasalingam Academy of Research and Education	
Class XII – State Board of Andhra Pradesh	2023
SR Junior College	912 / 1000
Class X – State Board of Andhra Pradesh	2021
Chinmaya Hari Hara Vidyalaya	541 / 600

Technical Skills

- **Programming:** Python, C (Embedded / Arduino)
- **Embedded / Hardware:** ESP32, ESP32-CAM, Arduino UNO, Sensors, Relay Modules, LoRa (SX1278)
- **Tools:** MATLAB, Simulink, GitHub, Arduino IDE, KiCad, VS Code
- **IoT / Cloud:** ThingSpeak, Supabase Realtime Database, Vercel
- **Web Technologies:** HTML, CSS

Projects

Python Automation Assistant (JARVIS-style CLI Tool) [GitHub]

Python, os / subprocess / SpeechRecognition libraries

- Developed a Python automation assistant for executing system commands, opening applications, and running scheduled tasks.
- Implemented modular scripts for file operations, process management, and task automation.
- Documented reusable Python workflows on GitHub.

Smart Irrigation Automation System [GitHub]

ESP32, Python, ThingSpeak, Arduino IDE

- Built an ESP32-based soil moisture sensing system with relay-controlled pump automation, reducing manual irrigation monitoring through automated relay-based control.
- Integrated ThingSpeak API for real-time sensor logging and web-based monitoring.
- Wrote Python scripts for ThingSpeak data parsing and moisture trend monitoring.

Elephant Intrusion Detection and Crop Protection System [GitHub]

ESP32-CAM, Arduino UNO, PIR / Ultrasonic Sensors, LoRa (SX1278), C (Embedded), KiCad

- Designed a perimeter intrusion detection system using PIR and ultrasonic sensors on Arduino UNO; integrated an ESP32-CAM module to capture still images on motion trigger for visual verification of intrusion events.
- Prototyped LoRa-based (SX1278) long-range alert transmission to relay detection events to a remote receiver node, suitable for farm boundaries without cellular coverage; integration logic simulated and partially tested.
- Designed a basic KiCad PCB schematic for sensor, relay, and wireless module integration.

Internship

IoT Intern Emertxe Information Technologies

Embedded Systems & IoT Training

- Developed embedded C programs on Arduino UNO for sensor interfacing and serial communication.
- Integrated multiple sensors with microcontroller circuits and practised hardware-software debugging workflows.

Achievements & Leadership

- **Google Gemini Student Ambassador** – Selected to support AI awareness activities on campus.
- **S Grade** in Signal Processing using Simulink and DSP for FPGA – highest grade awarded in the course.
- Participated in campus hackathons and IoT innovation events, advancing to Round 3 with A-grade technical evaluations.
- Served as an **NSS Board Member**, contributing to student and community engagement activities.

Languages

Telugu (Native) | English (Fluent) | Tamil (Basic)